Prompt Engineering Foundation

What did I learn?

This sprint established the Prompt Engineering Foundation for AIQEP by applying IBM Prompt Engineering concepts to real-world engineering and business use cases. Instead of treating prompt engineering as a theoretical topic, every concept was implemented, evaluated, and standardised into reusable prompt assets for AIQEP and AI Immanuel's Bakehouse.

1. Prompt Engineering Fundamentals

What is Prompt Engineering?

Prompt Engineering is the process of designing, testing, refining, and standardising instructions that guide a Generative AI model to produce accurate, consistent, and reusable outputs for a specific business or engineering objective.

Within AIQEP, prompt engineering is treated as an engineering discipline rather than simple question asking. Prompts are designed, evaluated, refined, and stored as reusable engineering assets.

What is a Prompt?

A prompt is the structured input provided to a Generative AI model to perform a defined task.

A well-designed prompt typically contains:

- Instructions
- Context
- Input Data
- Output Indicators

The quality of the prompt directly influences the quality, consistency, and usefulness of the generated output.

AIQEP Example

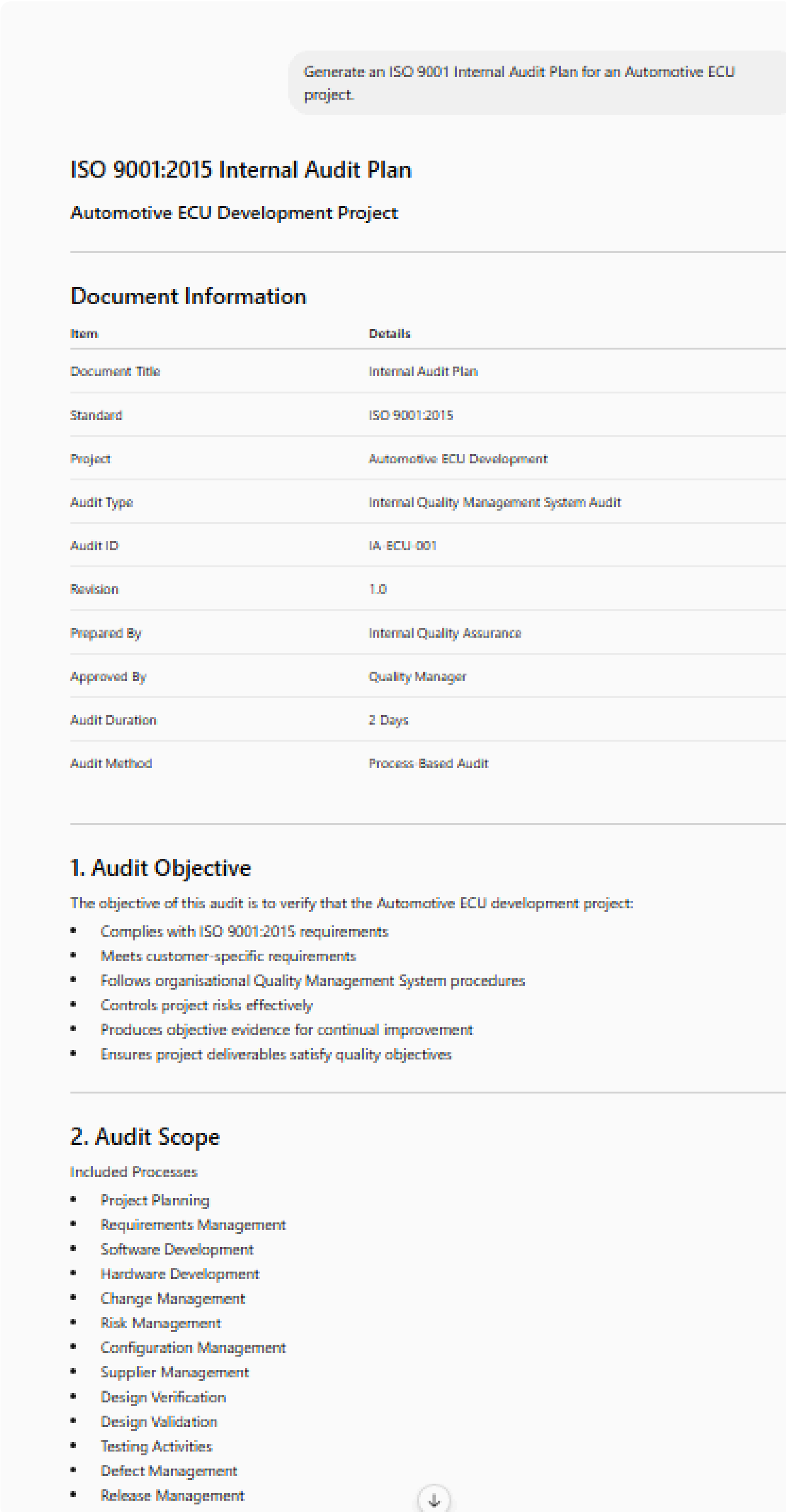
Prompt

> Generate an ISO 9001 Internal Audit Plan for an Automotive ECU project.

Expected Output

- Audit Scope
- Audit Objectives
- Audit Schedule
- Audit Checklist

- Risks
- Required Evidence



Generate an ISO 9001 Internal Audit Plan for an Automotive ECU project.

ISO 9001:2015 Internal Audit Plan

Automotive ECU Development Project

Document Information

Item	Details
Document Title	Internal Audit Plan
Standard	ISO 9001:2015
Project	Automotive ECU Development
Audit Type	Internal Quality Management System Audit
Audit ID	IA-ECU-001
Revision	1.0
Prepared By	Internal Quality Assurance
Approved By	Quality Manager
Audit Duration	2 Days
Audit Method	Process-Based Audit

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1. Audit Objective

The objective of this audit is to verify that the Automotive ECU development project:

- Complies with ISO 9001:2015 requirements
- Meets customer-specific requirements
- Follows organisational Quality Management System procedures
- Controls project risks effectively
- Produces objective evidence for continual improvement
- Ensures project deliverables satisfy quality objectives

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2. Audit Scope

Included Processes

- Project Planning
- Requirements Management
- Software Development
- Hardware Development
- Change Management
- Risk Management
- Configuration Management
- Supplier Management
- Design Verification
- Design Validation
- Testing Activities
- Defect Management
- Release Management

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Four Building Blocks of an Effective Prompt

Instead of using the IBM examples about global warming, replace them with your own examples.

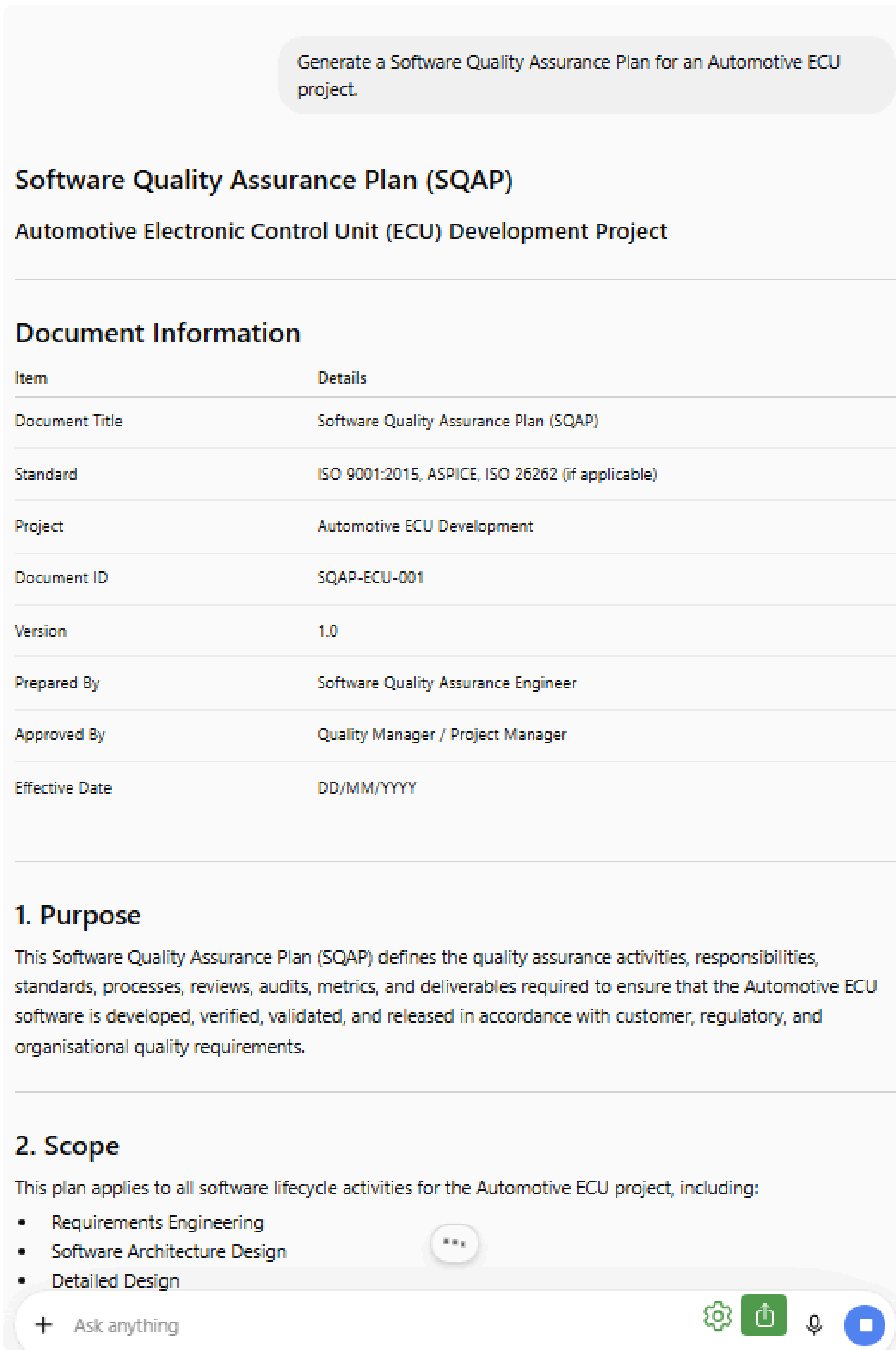
1. Instructions

Purpose

Clearly define what the AI should perform.

AIQEP Example

> Generate a Software Quality Assurance Plan for an Automotive ECU project.



Generate a Software Quality Assurance Plan for an Automotive ECU project.

Software Quality Assurance Plan (SQAP)

Automotive Electronic Control Unit (ECU) Development Project

Document Information

Item	Details
Document Title	Software Quality Assurance Plan (SQAP)
Standard	ISO 9001:2015, ASPICE, ISO 26262 (if applicable)
Project	Automotive ECU Development
Document ID	SQAP-ECU-001
Version	1.0
Prepared By	Software Quality Assurance Engineer
Approved By	Quality Manager / Project Manager
Effective Date	DD/MM/YYYY

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1. Purpose

This Software Quality Assurance Plan (SQAP) defines the quality assurance activities, responsibilities, standards, processes, reviews, audits, metrics, and deliverables required to ensure that the Automotive ECU software is developed, verified, validated, and released in accordance with customer, regulatory, and organisational quality requirements.

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2. Scope

This plan applies to all software lifecycle activities for the Automotive ECU project, including:

- Requirements Engineering
- Software Architecture Design
- Detailed Design

2. Context

Purpose

Provide background information so the AI understands the business or engineering scenario.

AIQEP Example

Project:

Automotive Body Control Module (BCM)

Development Phase:

System Integration Testing

Applicable Standard:

ASPICE Level 3

ISO 9001:2015

ISO 26262

Project:

Automotive Body Control Module (BCM)

Development Phase:

System Integration Testing

Applicable Standard:

ASPICE Level 3

ISO 9001:2015

ISO 26262

Show less

Software Quality Assurance Plan (SQAP)

Automotive Body Control Module (BCM)

Development Phase: System Integration Testing

Document Information

Item	Details
Document Title	Software Quality Assurance Plan (SQAP)
Project	Automotive Body Control Module (BCM)
Development Phase	System Integration Testing
Applicable Standards	ASPICE Level 3, ISO 9001:2015, ISO 26262
Document ID	SQAP-BCM-SIT-001
Version	1.0
Prepared By	Software Quality Assurance Engineer
Reviewed By	Test Manager / Project Manager
Approved By	Qu[illegible] Manager
Classification	Internal

Software Quality Assurance Plan (SQAP)

Automotive Body Control Module (BCM)

Development Phase: System Integration Testing

Document Information

Item	Details
Document Title	Software Quality Assurance Plan (SQAP)
Project	Automotive Body Control Module (BCM)
Development Phase	System Integration Testing
Applicable Standards	ASPICE Level 3, ISO 9001:2015, ISO 26262
Document ID	SQAP-BCM-SIT-001
Version	1.0
Prepared By	Software Quality Assurance Engineer
Reviewed By	Test Manager / Project Manager
Approved By	Quality Manager
Classification	Internal

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1. Purpose

This Software Quality Assurance Plan defines the quality assurance activities, controls, responsibilities, reviews, audits, metrics, and deliverables required during the **System Integration Testing (SIT)** phase of the Automotive Body Control Module (BCM) project.

The objective is to ensure that the BCM software satisfies customer, safety, regulatory, and organisational requirements while complying with:

- ASPICE Level 3 Process Capability
- ISO 9001:2015 Quality Management System
- ISO 26262 Functional Safety

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2. Scope

This SQAP covers all quality assurance activities associated with System Integration Testing, including:

- Test Planning
- Test Environment Qualification
- Test Case Review
- Test Execution
- Requirements Traceability
- Functional Testing
- Regression Testing
- Interface Testing
- Network Communication Testing
- Defect Management
- Configuration Management
- Safety Requirement Verification
- Quality Audits
- Test Reporting
- Release Readiness Assessment

Ask anything

3. Input Data

Purpose

Supply the information the AI will use to generate its response.

AIQEP Example

Field	Value

Customer	OEM ABC
Platform	AUTOSAR
ASIL	B
Phase	Validation
Known Issues	Headlamp delay, CAN timeout, Diagnostic failure

Field	Value
Customer	OEM ABC
Platform	AUTOSAR
ASIL	B
Phase	Validation
Known Issues	Headlamp delay, CAN timeout, Diagnostic failure

Based on the project information in the image, here is the extracted project context that can be incorporated into the Software Quality Assurance Plan.

Field	Value
Project	Automotive Body Control Module (BCM)
Customer	OEM ABC
Platform	AUTOSAR
Safety Integrity Level	ASIL B
Current Development Phase	Validation
Known Issues	Headlamp delay, CAN timeout, Diagnostic failure

Using these details, the project-specific sections of the SQAP would be updated as follows.

Project Overview

Project Name

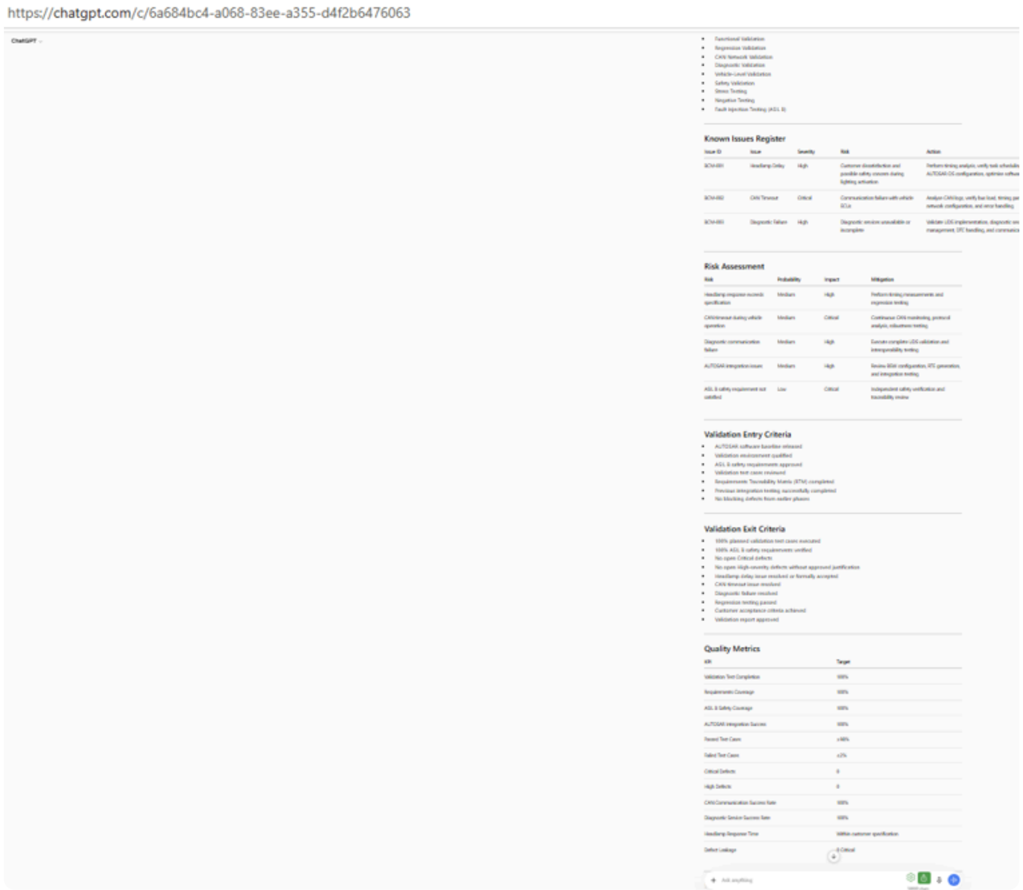
Automotive Body Control Module (BCM)

Customer

OEM ABC

Software Platform

AUTOSAR Classic Platform



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ChatGPT

- Functional Validation
- Regression Validation
- CAN Network Validation
- Diagnostic Validation
- Vehicle-Level Validation
- Safety Validation
- Stress Testing
- Negative Testing
- Fault Injection Testing (ASIL B)

Known Issues Register

Issue ID	Issue	Severity	Risk	Action
BCM-001	Headlamp Delay	High	Customer dissatisfaction and possible safety concern during lighting activation	Perform timing analysis, verify task schedulin… AUTOSAR OS configuration, optimize softwa…
BCM-002	CAN Timeout	Critical	Communication failure with vehicle ECUs	Analyze CAN logs, verify bus load, timing par… network configuration, and error handling
BCM-003	Diagnostic Failure	High	Diagnostic services unavailable or incomplete	Validate UDS implementation, diagnostic ses… management, DTC handling, and communicatio…

Risk Assessment

Risk	Probability	Impact	Mitigation
Headlamp response exceeds specification	Medium	High	Perform timing measurements and regression testing
CAN timeout during vehicle operation	Medium	Critical	Continuous CAN monitoring, protocol analysis, robustness testing
Diagnostic communication failure	Medium	High	Execute complete UDS validation and interoperability testing
AUTOSAR integration issues	Medium	High	Review BSW configuration, RTE generation, and integration testing
ASIL B safety requirement not satisfied	Low	Critical	Independent safety verification and traceability review

Validation Entry Criteria

- AUTOSAR software baseline released
- Validation environment qualified
- ASIL B safety requirements approved
- Validation test cases reviewed
- Requirements Traceability Matrix (RTM) completed
- Previous integration testing successfully completed
- No blocking defects from earlier phases

Validation Exit Criteria

- 100% planned validation test cases executed
- 100% ASIL B safety requirements verified
- No open Critical defects
- No open High-severity defects without approved justification
- Headlamp delay issue resolved or formally accepted
- CAN timeout issue resolved
- Diagnostic failure resolved
- Regression testing passed
- Customer acceptance criteria achieved
- Validation report approved

Quality Metrics

KPI	Target
Validation Test Completion	100%
Requirements Coverage	100%
ASIL B Safety Coverage	100%
AUTOSAR Integration Success	100%
Passed Test Cases	≥98%
Failed Test Cases	≤2%
Critical Defects	0
High Defects	0
CAN Communication Success Rate	100%
Diagnostic Service Success Rate	100%
Headlamp Response Time	Within customer specification
Defect Leakage	0 Critical

Ask anything

4. Output Indicators

Purpose

Specify the expected format and quality of the response.

AIQEP Example

Generate:

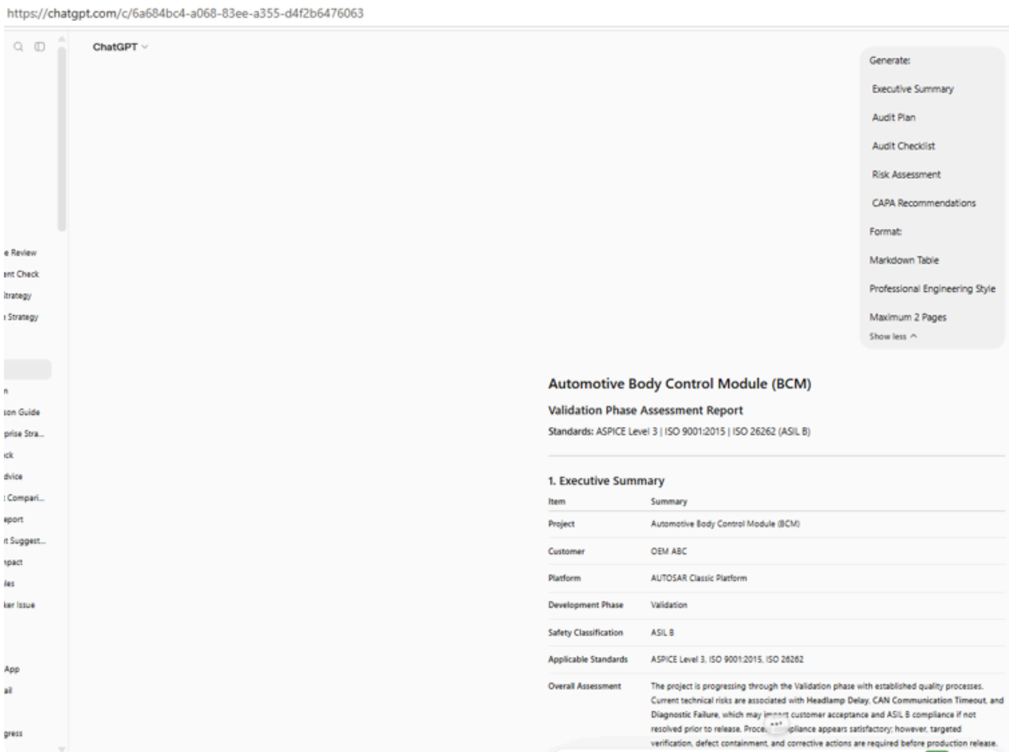
- Executive Summary
- Audit Plan
- Audit Checklist
- Risk Assessment
- CAPA Recommendations

Format:

Markdown Table

Professional Engineering Style

Maximum 2 Pages



Automotive Body Control Module (BCM)

Validation Phase Assessment Report

Standards: ASPICE Level 3 | ISO 9001:2015 | ISO 26262 (ASIL B)

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1. Executive Summary

Item	Summary
Project	Automotive Body Control Module (BCM)
Customer	OEM ABC
Platform	AUTOSAR Classic Platform
Development Phase	Validation
Safety Classification	ASIL B
Applicable Standards	ASPICE Level 3, ISO 9001:2015, ISO 26262
Overall Assessment	The project is progressing through the Validation phase with established quality processes. Current technical risks are associated with **Headlamp Delay**, **CAN Communication Timeout**, and **Diagnostic Failure**, which may impact customer acceptance and ASIL B compliance if not resolved prior to release. Process compliance appears satisfactory; however, targeted verification, defect containment, and corrective actions are required before production release.
Overall Risk Rating	**Medium–High**
Recommendation	Continue validation with daily defect review, enhanced regression testing, safety verification, and management oversight until all critical quality objectives are achieved.

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2. Audit Plan

Audit Area	Standard Reference	Objective	Evidence	Owner	Priority
Project Planning	ISO 9001 Cl.6, ASPICE MAN.3	Verify planning and milestone control	Project Plan, Schedule	Project Manager	Medium
Requirements Traceability	ASPICE SYS.2, SYS.4	Confirm complete bidirectional traceability	RTM, Polarion/DOORS	Systems Engineer	High
Validation Testing	ASPICE SYS.5, SWE.6	Verify execution and coverage	Test Reports, Logs	Test Lead	High
Functional Safety	ISO 26262	Verify ASIL B safety requirements	Safety Case, Safety Reports	Functional Safety Manager	Critical
Configuration Management	ASPICE SUP.8	Verify baseline integrity	Git Baseline, Release Records	Configuration Manager	High
Defect Management	ASPICE SUP.9	Verify defect lifecycle effectiveness	Jira/ALM Reports	QA Lead	High
Diagnostic Validation	ISO 14229 / Customer Requirements	Verify UDS diagnostics	Diagnostic Logs	Validation Engineer	High
Management Review	ISO 9001 Cl.9.3	Verify quality performance review	Review Minutes	Quality Manager	Medium

3. Audit Checklist

Audit Question	Evidence Required	Expected Result	Status
Are validation test cases approved?	Test Plan, Review Records	All approved	☐
Is requirements traceability complete?	RTM	100% traceability	☐
Are ASIL B safety requirements validated?	Safety Test Report	Complete verification	☐
Has AUTOSAR integration been verified?	Integration Report	No integration issues	☐
Are CAN communication timeouts resolved?	CAN Logs	No timeout events	☐
Is headlamp response within specification?	Timing Report	Meets OEM requirement	☐
Are diagnostic services operating correctly?	UDS Test Report	All services passed	☐
Are configuration baselines controlled?	CM Records	Approved baseline	☐
Are all critical defects closed?	Defect Dashboard	Zero Critical	☐
Has regression testing been completed?	Regression Report	≥98% pass rate	☐
Has release readiness been reviewed?	Release Checklist	Approved	☐
Are internal quality audits completed?	Audit Report	No Major NCRs	☐

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4. Risk Assessment

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation
R-01	Headlamp delay exceeds OEM specification	Medium	High	**High**	Perform timing analysis, optimise scheduling, execute regression tests
R-02	CAN timeout during ECU communication	Medium	Critical	**Critical**	Analyse CAN traffic, validate network timing, stress test communication
R-03	Diagnostic (UDS) service failure	Medium	High	**High**	Validate all diagnostic services, verify DTC handling and session management
R-04	Incomplete ASIL B verification	Low	Critical	**High**	Complete safety verification, independent review and traceability audit
R-05	Configuration baseline mismatch	Low	High	**Medium**	Perform configuration audit before release
R-06	Regression defects introduced after fixes	Medium	High	**High**	Execute full regression suite before release approval

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5. CAPA Recommendations

Finding	Root Cause (Likely)	Corrective Action	Preventive Action	Owner	Target
Headlamp delay	Software scheduling/configuration	Optimise task execution and timing	Add response-time KPI to regression suite	Software Lead	Before Release
CAN timeout	Network configuration or timing	Correct communication parameters and verify under load	Introduce automated CAN stress testing	Integration Lead	Before Release
Diagnostic failure	Incomplete UDS implementation or configuration	Correct diagnostic services and revalidate	Expand diagnostic regression coverage	Validation Lead	Before Release
Safety verification gaps	Missing verification evidence	Complete ASIL B verification and update safety case	Independent functional safety reviews	Safety Manager	Before SOP
Defect recurrence	Weak root cause analysis	Apply structured RCA (5 Whys/Fishbone) for recurring defects	Periodic CAPA effectiveness reviews	QA Manager	Ongoing
Configuration inconsistencies	Baseline control weakness	Re-establish validated software baseline	Mandatory configuration audit before every release	Configuration Manager	Ongoing

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Overall Audit Conclusion

Category	Assessment
Process Compliance	**Compliant with minor improvements required**
ASPICE Level 3 Readiness	**On Track**
ISO 9001 Compliance	**Satisfactory**
ISO 26262 (ASIL B) Compliance	**Pending closure of validation evidence and safety-related defects**
Release Recommendation	**Conditional Approval** – Release only after closure of **Headlamp Delay**, **CAN Timeout**, **Diagnostic Failure**, successful regression testing, and completion of ASIL B verification evidence.

Engineering Insight

> **Engineering Insight:** AIQEP standardises every reusable prompt using the four building blocks: Instructions, Context, Input Data, and Output Indicators. This approach improves response consistency, reduces ambiguity, and enables prompt reuse across quality engineering and bakery use cases.

2. Prompt Life Cycle

AIQEP adopts a PDCA-based Prompt Engineering Lifecycle. Each prompt is planned, executed, evaluated, refined, and finally standardised before being added to the reusable prompt library. This approach aligns prompt engineering with continuous improvement principles used in quality management.

```
                 AIQEP Prompt Engineering Framework

            PLAN

 ┌──────────────────────────┐
 │ Understand Problem       │
 │ Define Objective         │
 │ Choose Prompt Pattern    │
 └────────────┬─────────────┘
              │
              ▼
           DO

 ┌──────────────────────────┐
 │ Create Initial Prompt    │
 │ Execute Prompt           │
 └────────────┬─────────────┘
              │
              ▼
          CHECK

 ┌──────────────────────────┐
 │ Evaluate Response        │
 │ Accuracy                 │
 │ Completeness             │
 │ Consistency              │
 └────────────┬─────────────┘
              │
              ▼
             ACT

 ┌──────────────────────────┐
 │ Refine Prompt            │
 │ Standardize              │
 │ Publish                  │
 │ Reuse                    │
 └──────────────────────────┘
```

3. Best Practices

Best Practice	Why it Matters
Clearly define the objective	Reduces ambiguity
Provide relevant context	Improves response quality
Supply structured input	Produces more accurate outputs

Define output expectations	Creates consistent results
Test and refine prompts	Improves reliability
Store successful prompts	Enables reuse

4. IBM Module 1 Summary

Topic	Application in AIQEP
Prompt Fundamentals	Standardised prompt structure
Four Building Blocks	Used in all reusable prompts
Prompt Lifecycle	PDCA-based engineering workflow
Prompt Refinement	Iterative improvement process
Best Practices	Prompt standards for AIQEP

5. AI QEP Examples

Asset	Purpose
Software QA Plan Generator	Generate project quality plans
Internal Audit Plan Generator	Produce ISO 9001 audit plans
CAPA Generator	Generate corrective action reports
FMEA Generator	Support risk analysis
Fishbone Generator	Root cause analysis
Risk Assessment Assistant	Identify project risks

6. Bakehouse Examples

Asset	Purpose
Recipe Generator	Standardise recipes
Product Development Assistant	Create new products
Menu Planning Assistant	Build menus

Marketing Generator	Campaign creation
Packaging Assistant	Packaging ideas
Product Photography	Image prompt library

This supports AIQEP-10 and part of AIQEP-15.